Transcriptional analysis of molecular markers related to the pathogenesis of Parkinson's disease at different stages of neuronal differentiation

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Introduction

Parkinson's disease (PD) is a neurodegenerative disorder that primarily affects dopaminergic neurons. PD is common in people up to 50 years of age, but neuronal degeneration begins many years before the first symptoms. However, the point of onset of the disease is currently debated.

Investigation of pathological mechanisms has revealed to alpha-synuclein (α-syn) as a key molecule in the development of PD. In this regard, it is now known that overexpression of α-syn results in increased susceptibility to oxidative stress, as observed in a study with dopaminergic neurons differentiated from pluripotent stem cells induced with a triplication of the a-syn gene (hiPSC 3XSNCA) (Byers et al., 2011). Interestingly, there were similar results in neuronal precursors derived from hiPSC 3X*SNCA* (Flierl et al., 2014). Thus, it appears that α-syn may induce changes leading to the development of PD before the mature stage of dopaminergic neurons.

With that in mind, we set out to identify genes and pathways dysregulated across different stages of neuronal maturation as possible early markers of PD.

Methods

Transcriptional data were obtained from Devine et al., (2011), whose data are available in the GEO repository under accession code GSE28365. The data set contain transcriptional data from cells with the α-syn triplication (3XSNCA) at different maturation stages. The data analysis workflow is shown in Fig.1.

To select the differentially expressed genes (DEGs) we established a p-value ≤ 0.05 and |logFC| ≥ 1.0 as the cutoff point. To identify the processes and pathways most affected by DEGs, we ranked the enriched gene ontology-biological processes (GO-BP) and KEGG pathways according to the p-value. Finally, we selected the top 5 of genes with the highest degree of interaction within the interaction networks.

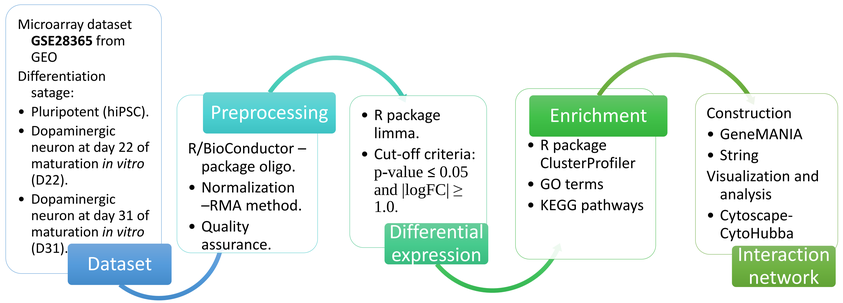

Fig.1 Data analysis workflow

RMA:Robust multichip average; GO: Gene ontology; KEGG: Kyoto Encyclopedia of Genes and Genomes.

Results

We found 131 DEGs in pluripotent cells, 111 DEGs in dopaminergic neurons at D22, and 231 DEGs in dopaminergic neurons at D31. When we performed the enrichment analysis of those DEGs (Fig.2 A-C) we found that development-related biological processes were enriched in pluripotent stage. In addition, some biological processes related to lipid metabolism and synapsis were enriched in dopaminergic neurons at D22. Finally, biological processes related to cell movement were enriched in dopaminergic neurons at D31.

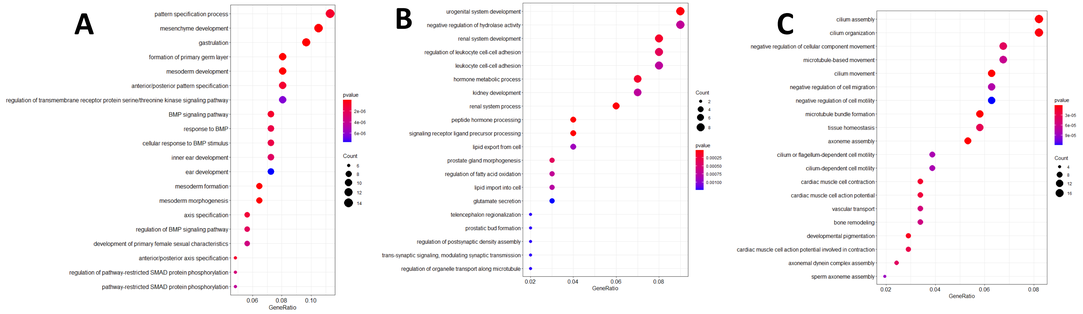

Fig. 2 Top 20 of GO-Biological processes enriched in three different stages.

A. Enrichment analysis in pluripotent stage. **B.** Enrichment analysis in dopaminergic neurons at D22. **C.** Enrichment analysis in dopaminergic neurons at D31.

Furthermore, KEGG pathway analysis showed an enrichment in pathways involved in neuronal differentiation in both pluripotent and dopaminergic neurons at D31 stages (Table 1 and Table 2).

Table 1. KEGG pathways enriched in pluripotent stage.

Pathway ID	Name	p-value	Genes
Hsa04550	Signaling pathways regulating pluripotency of stem cells	0.00018	BMP4 / HAND1 / INHBA /INHBE / NODAL / WNT3 /WNT8A
Hsa04310	Wnt signaling pathway	0.00045	CER1/ DKK4/ LEF1/ LGR5/ SFRP1/ WNT3 / WNT8A
hsa04350	TGF-beta signaling pathway	0.00111	BMP4 / FST /INHBA / INHBE / NODAL

Table 2. KEGG pathways enriched in dopaminergic neuron D31.

Pathway ID	Name	p-value	Genes
hsa04310	Wnt signaling pathway	0.000529077	DAAM2/ FZD4 / LGR5 /NKD1 / SERPINF1 / SFRP2 /SFRP5 / WIF1

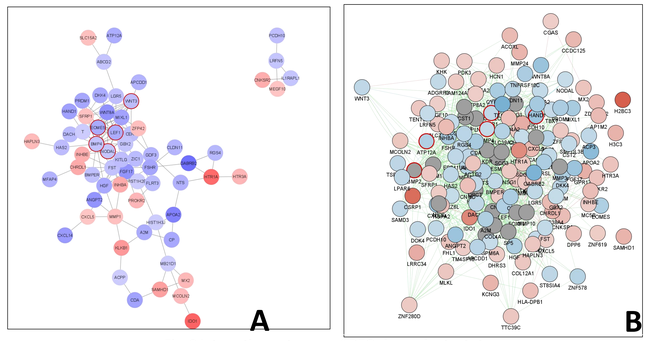

Fig. 3 Interaction networks of DEGs in pluripotent stage.

A. Interaction network constructed by String. **B.** Interaction network constructed by GeneMania.

The genes with more interactions are highlighted with color red in the interaction networks (Fig. 3-5).

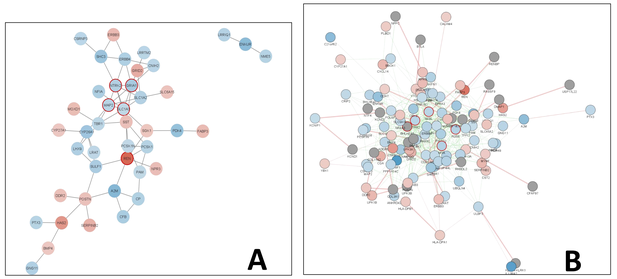

Fig. 4 Interaction networks of DEGs in dopaminergic neuron at D22.

A. Interaction network constructed by String. **B.** Interaction network constructed by GeneMania.

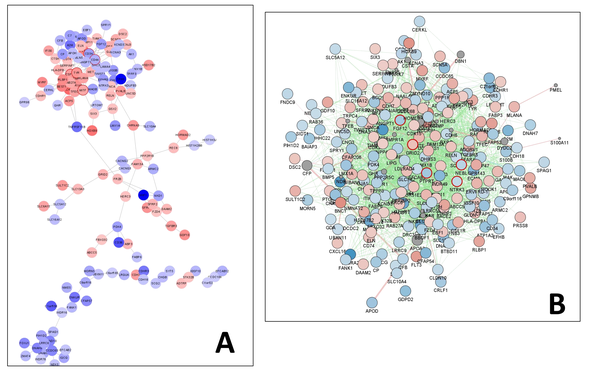

Fig. 5 Interaction networks of DEGs in dopaminergic neuron at D31.

A. Interaction network constructed by String. **B.** Interaction network constructed by GeneMania.

Conclusions

These results suggest that α-sin triplication could induce alterations in the biological processes involved in neuronal differentiation at different stages of maturation *in vitro*. Furthermore, it appears that altered expression of some genes could be involved in PD pathology development. However, further studies on the role of those genes are needed.

Acknowledgements

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References

Byers, B., Cord, B., Nguyen, H., Schüle, B., Fenno, L., Lee, P., & Palmer, T. (2011). SNCA Triplication Parkinson's Patient's iPSC-derived DA Neurons Accumulate a-Synuclein and Are Susceptible to Oxidative Stress. PLoS ONE, 6(11). https://doi.org/10.1371/journal.pone.0026159

Devine, M. J., Ryten, M., Vodicka, P., Thomson, A. J., Burdon, T., Houlden, H., Cavaleri, F., Nagano, M., Drummond, N. J., Taanman, J.-W., Schapira, A. H., Gwinn, K., Hardy, J., Lewis, P. A., & Kunath, T. (2011). Parkinson's disease induced pluripotent stem cells with triplication of the α-synuclein locus. Nature Communications, 2. https://doi.org/https://doi.org/10.1038/ncomms1453

Flierl, A., Oliveira, L., Falomir-Lockhart, L., Mak, S., Hesley, J., Soldner, F., .. ., & Schüle, B. (2014). Higher Vulnerability and Stress Sensitivity of Neuronal Precursor Cells Carrying an Alpha-Synuclein Gene Triplication. PLoS ONE. https://doi.org/10.1371/journal.pone.0112413